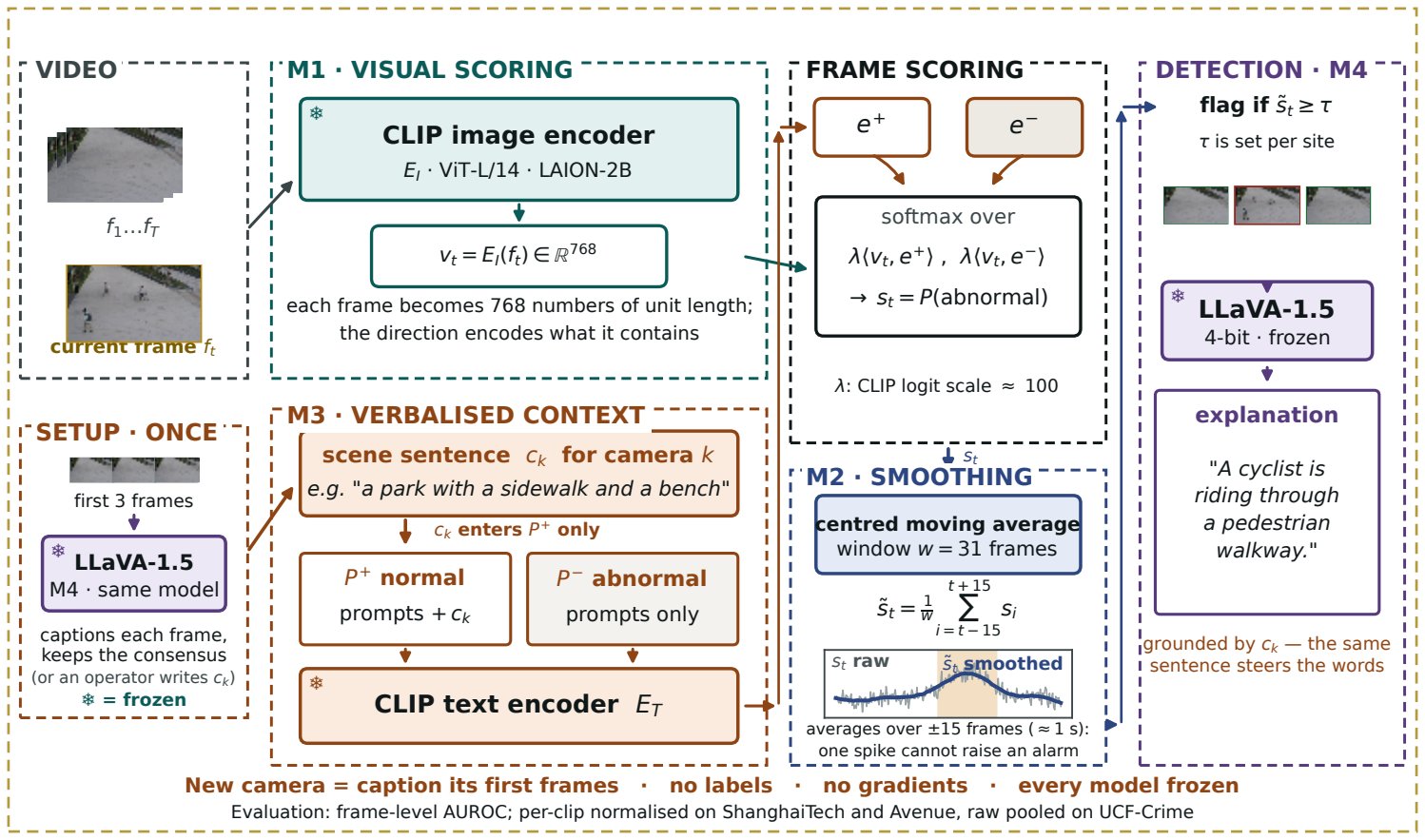

VIDEO
$f_1 \dots f_T$
current frame f_t
SETUP · ONCE
first 3 frames
LLaVA-1.5
M4 · same model
captions each frame, keeps the consensus
(or an operator writes c_k)
❄ = frozen
M1 · VISUAL SCORING
❄
CLIP image encoder
E_I · ViT-L/14 · LAION-2B
$v_t = E_I(f_t) \in \mathbb{R}^{768}$
each frame becomes 768 numbers of unit length;
the direction encodes what it contains
M3 · VERBALISED CONTEXT
scene sentence c_k for camera k
e.g. "a park with a sidewalk and a bench"
c_k enters P^+ only
P^+ normal
prompts + c_k
P^- abnormal
prompts only
❄
CLIP text encoder E_T
FRAME SCORING
e^+
e^-
softmax over
$\lambda\langle v_t, e^+\rangle$, $\lambda\langle v_t, e^-\rangle$
$\rightarrow s_t = P(\text{abnormal})$
λ: CLIP logit scale $\approx$ 100
s_t
M2 · SMOOTHING
centred moving average
window $w = 31$ frames
$\tilde{s}_t = \frac{1}{w} \sum_{i=t-15}^{t+15} s_i$
s_t raw
$\tilde{s}_t$ smoothed
averages over ±15 frames (≈ 1 s):
one spike cannot raise an alarm
DETECTION · M4
flag if $\tilde{s}_t \geq \tau$
τ is set per site
❄
LLaVA-1.5
4-bit · frozen
explanation
"A cyclist is riding through a pedestrian walkway."
grounded by c_k — the same sentence steers the words
New camera = caption its first frames · no labels · no gradients · every model frozen
Evaluation: frame-level AUROC; per-clip normalised on ShanghaiTech and Avenue, raw pooled on UCF-Crime